
Part B**Ex: 11****Title:** Event Budget Allocation

Problem Description: An event planner is organizing a conference and has a fixed budget to allocate across various categories such as venue, catering, speakers, and marketing. The planner needs to ensure that the total expenditure does not exceed the budget while maximizing the quality of the event. Set of Expenses (with costs): Venue rental-\$5000, Catering \$2000, Keynote Speaker-\$3000, Marketing-\$1500, Audio/Visual Equipment-\$1000. Given Fixed budget-\$8000

Method: Use backtracking technique to solve the problem.

Theory Reference: Module 4

Algorithm**1) Mark Current Element as Selected:**

Set $x[k] = 1$.

2) Check if Adding $w[k]$ to Current Sum Matches Target:

If $s + w[k] == d$, then:

- A valid subset is found.
- Increment and print the subset count.
- Print all elements i where $x[i] == 1$ (i.e., selected elements and their categories).

3) Left Subtree (Include Current Element):

If $k + 1$ is a valid index AND $s + w[k] + w[k+1] \leq d$, then:

- **Recur:** `sum_of_subsets(s + w[k], k + 1, rem - w[k])`
- This explores the case where the current item is included.

4) Right Subtree (Exclude Current Element):

If $k + 1$ is a valid index AND $(s + rem - w[k] \geq d)$ AND $s + w[k+1] \leq d$, then:

- Set $x[k] = 0$ (exclude current element).
 - **Recur:** `sum_of_subsets(s, k + 1, rem - w[k])`
 - This explores the case where the current item is excluded.
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Program

```
import java.util.Scanner;

public class SubsetSum
{
    static int x[],w[],count,d;

    static String[] categories ;

    static void sum_of_subsets(int s, int k, int rem)
    {
        x[k] = 1;

        if( s + w[k] == d)
        {
            //if sub set found

            System.out.println("Subset/Event Planning = "+ ++count);

            for(int i=0 ; i <= k ; i++)

                if ( x[i] == 1)

                    System.out.println(categories[i]+" : "+w[i]);

            System.out.println();
        }

        else if ( s + w[k] + w[k+1] <= d)//left tree evaluation

            sum_of_subsets(s+w[k], k+1, rem-w[k]);

        if( ( s + rem - w[k] >= d) && ( s + w[k+1]) <= d)//right tree evaluation
        {
```

```
        x[k] = 0;

        sum_of_subsets(s,k+1,rem-w[k]);

    }

}

public static void main(String[] args)

{

    int sum = 0,n;

    Scanner s=new Scanner(System.in);

    System.out.println("Enter no of categories");

    n = s.nextInt();

    w = new int[n];

    x = new int[n];

    categories=new String[n];

    s.nextLine();

    System.out.println("Enter the category names followed by their fixed budgets (in
increasing order of budget:");

    for (int i = 0; i < n; i++)

    {

        System.out.print("Category " + (i + 1) + " name: ");

        categories[i] = s.nextLine();

        System.out.print("Category " + (i + 1) + " budget: ");

        w[i] = Integer.parseInt(s.nextLine()); // Use nextLine to avoid newline issues

        sum += w[i];

    }

}
```

```

    }

    System.out.println("Enter the budget to host the event");

    d = s.nextInt();

    if ( ( sum < d ) || ( w[0] > d ) )

        System.out.println("No subset possible-Unable to host the events");

    else

        sum_of_subsets(0,0,sum);

    }

}

```

Output

```

run:
Enter no of categories
5
Enter the category names followed by their fixed budgets (in increasing order of budget):
Category 1 name: Audio/Visual equipment
Category 1 budget: 1000
Category 2 name: Marketing
Category 2 budget: 1500
Category 3 name: Catering
Category 3 budget: 2000
Category 4 name: Speaker
Category 4 budget: 3000
Category 5 name: Venue Rental
Category 5 budget: 5000
Enter the budget to host the event
8000
Subset/Event Planning - 1
Audio/Visual equipment : 1000
Catering : 2000
Venue Rental : 5000

Subset/Event Planning - 2
Speaker : 3000
Venue Rental : 5000

BUILD SUCCESSFUL (total time: 1 minute 43 seconds)

run:
Enter no of categories
4
Enter the category names followed by their fixed budgets (in increasing order of budget):
Category 1 name: Venue Rental
Category 1 budget: 4000
Category 2 name: Marketing
Category 2 budget: 5000
Category 3 name: Catering
Category 3 budget: 6000
Category 4 name: Speaker
Category 4 budget: 7000
Enter the budget to host the event
3000
No subset possible-Unable to host the events
BUILD SUCCESSFUL (total time: 44 seconds)

```
